

Divit Mittal

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WORK EXPERIENCE

UbiOps (Yes!Delft)

City: Delft | Country: Netherlands

[15/12/2021 - 28/02/2022]

DevOps & System Administrator

Administered containerized environments using Docker/Kubernetes for multi-tenant ML workloads, implemented CI/CD pipelines for automated model deployment, and maintained cloud infrastructure across AWS/GCP.

Remote

[01/07/2024 - Current]

Open Source & Community Contributions

Active contributor to the functional programming ecosystem, particularly within the Nix community. Participated in Nix consulting bounties (Summer 2024) with **Tweag**, **ModusCreate**, and **Blockfrost**, providing expertise on complex NixOS configurations, package derivations, and infrastructure automation for enterprise clients.

EDUCATION & TRAINING

[10/09/2022 - Current]

Bachelor of Technology in Data Science & Engineering

Manipal University <https://jaipur.manipal.edu/>

City: Jaipur, Rajasthan | Country: India | National classification: NCC5

[01/09/2021 - 01/09/2022]

Bachelor of Science in Computer Science & Engineering

Delft University of Technology tudelft.nl

City: Delft | Country: Netherlands | Final grade: 8.0 | Type of credits: ECTS | Number of credits: 20

[01/08/2020 - 28/08/2020]

Bachelor of Computing in Computer Science

National University of Singapore <https://www.nus.edu.sg/>

[31/03/2018 - 12/07/2020]

All India Senior School Certificate Examination

Central Board of Secondary Education (CBSE)

Address: Delhi Public School, R.K. Puram, New Delhi (India) | Final grade: A1 : ~96% | National classification: NSQF level 4 - (NCC 2)

Diploma: <https://europa.eu/!yp79Jh>

Transcript : <https://europa.eu/!BB98Ww>

Character Certificate: <https://europa.eu/!HD84fw>

[31/03/2016 - 28/05/2018]

All India Secondary School Examination

Central Board of Secondary Education (CBSE)

Address: Bhatnagar International School, Bhera Enclave Paschim Vihar, New Delhi (India) | Final grade: A1 : ~97% | National classification: NSQF level 2 - (NCWP 2)

Diploma: <https://europa.eu/!Mv98jr>

Transcript: <https://europa.eu/!uG83GN>

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

German/Deutsch (Goethe-Zertifikat B2)

LISTENING: B2 **READING:** B2 **WRITING:** B2

SPOKEN PRODUCTION: B2 **SPOKEN INTERACTION:** B2

English (IELTS)

LISTENING: C2 **READING:** C2 **WRITING:** C1

SPOKEN PRODUCTION: C1 **SPOKEN INTERACTION:** C2

SKILLS

Machine Learning & Data Science (Mathematics)

Deep Learning: Keras & PyTorch | Machine Learning: scikit-learn, tensorflow | LLM: Transformers, RAG, Encoders, & Decoders | Big Data: Hadoop, Spark, Hive, Kafka | Computer Vision: OpenCV, ITK

DevOps & Linux SysAdmin

NixOS Expert | IaC: nix, ansible & terraform | VCS: Git & Hg | Cloud Experience: AWS, GCP & Firebase | Shells: bash, zsh, fish, pwsh | Virtualization & Containerization: Proxmox, KVM, Docker & Podman | CI/CD: GitHub Actions & Jenkins

PROJECTS

[04/2024 - Current]

OS-nixCfg: A Personal Infrastructure as Code configuration

Nix, nix-darwin, **NixOS**, home-manager, etc. configurations to attain my **deterministically reproducible** OS, written in the **functional** "Nix" expression language. Maintains:

- Darwin-kernel systems, i.e. macOS
- Android systems
- Desktop Linux-kernel systems, e.g. NixOS, Arch Linux, RHEL
- WSL2 on Windows

Link: <https://github.com/DivitMittal/OS-nixCfg>

[02/2024]

Programming Languages Documentation Conversational AI @ IEEE MUJ

RAG-enhanced **LLaMa-like transformer** neural network based LLM for assistance on programming languages/frameworks' documentations, with a **streamlit** front-end. Everything was implemented from scratch:

- Pre-trained **LLaMa-like transformer** with capability from **tokenization** to **embedding** (positional & word2vector) to output probabilities.
- **RAG** was implemented via **Qdrant** vector database on a **DigitalOcean Docker** container.

Link: <https://github.com/DivitMittal/DocAssist-LLM>

[10/2023]

Generation of "Hazard Map" for safely navigating a lunar lander

Is implemented by utilising the sensory data from the **ISRO**'s Chandrayaan-2 TMC2 & OHRC payloads for **SIH** (Smart India Hackathon). The project entails wrangling multiple data in form of multispectral high-res. imagery & stereo-triplet panchromatic data of lunar terrain. These multiple streams of data are fused together and analysed respectively for ground truth via **SRGANs** & **DCNNs**.

HONOURS AND AWARDS

[01/09/2024]

BobbleAI Datathon Winner **Awarding institution:** BobbleAI Keyboard, Gurgaon, Haryana

Processed raw product data, i.e. [BobbleAI Keyboard](#) (7Cr+ users) event records in the last 5 days before July 1, 2024, i.e., about 8 million rows. Extracted & transformed JSON, to preprocess, ML-filter and store as structured **Apache Parquet** on **AWS S3** via serverless **AWS Lambda** functions & **PySpark** on **AWS Glue**.

[01/10/2023]

Proglint's Computer Vision 2k23 National Hackathon 5th Position Holder **Awarding institution:** Alliance University, Anekal, Karnataka

- Leverage the high-fidelity **CARLA** simulator to create diverse driving scenarios for an **autonomous vehicle driving system** with RGB & LiDAR sensors. Utilizing deep learning techniques, i.e. **CNNs** for **semantic segmentation** to improve the vehicle's understanding of its surroundings, & manoeuvre.
- A **Driver Drowsiness Detection** system that uses **Siamese Neural Networks** and computer vision techniques, like **haar cascade** wave filters to monitor a driver's facial features in real-time and detect signs of fatigue. By analyzing eye states, & mouth movements, the system identifies potential drowsiness and enhances road safety.

Link: <https://github.com/DivitMittal/CARLA-Autonomous-Driving>

[01/01/2020]

International Rank 7 at the International Mathematics Olympiad **Awarding institution:** The Science Olympiad Foundation, New Delhi

- [01/12/2019] **International Rank 49 at the International English Olympiad (IEO)** Awarding institution: The British Council Division, British High Commission, New Delhi
- [01/01/2019] **India National Level Runner-up at the Vidyarthi Vigyan Manthan** Awarding institution: Vijnana Bharati (VIBHA), with Vigyan Prasara, under the Department of Science and Technology & NCERT